

Prediction Markets as an Information Layer

Part 2 of 6: When Market Structure Becomes Meaning

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Abstract

If prediction markets serve as an information layer, then market design is not just plumbing, it's part of what the price actually means. This paper will argue that different prediction market platforms produce meaningfully different signals, because their structures encode different assumptions about who can trade, what risks they can take, and how disputes get resolved. I will use Kalshi and Polymarket as contrasting case studies and draw on both empirical data from the 2024 election as well as direct experience building on top of both platforms, to look at how liquidity, friction, and resolution trust shape the nature of what prices can tell us. My goal here is not to pick a winner. It's to explain why their prices shouldn't be treated as interchangeable, and why the gap between them is often the most interesting signal of all.

I. A Price Is Not Just a Number

If you've spent any time in prediction markets, you've probably had this thought:

A contract at 65 cents is a contract at 65 cents. The platform shouldn't matter. Belief is belief.

I used to think this too. After watching these markets since 2023 and building trading infrastructure on top of them, I've come to see it differently.

A market price doesn't emerge from nowhere. It's the output of a process; one shaped by who's allowed to trade, how much it costs to express a view, how resolution works, how much size participants can actually put on. A probability isn't just belief. It's belief that made it through a particular market's friction.

Once you see it that way, the idea that all prediction market prices are interchangeable falls apart. Different architectures filter different noise. They reward different kinds of conviction. And they produce signals that need to be read differently.

II. Two Very Different Machines

Kalshi and Polymarket get lumped together constantly. In practice, they're built on completely different assumptions about how markets should work.

Kalshi is a CFTC-regulated exchange. You need to pass compliance. Markets get created through a formal process. Resolution comes from internal teams using sources like the AP or official government data. The whole thing is designed to feel like traditional finance slow, deliberate, institutionally legible.

Polymarket is crypto-native. You connect a wallet, deposit USDC, and you're trading. Markets spin up fast. Resolution runs through UMA's Optimistic Oracle, outcomes are proposed onchain, assumed correct unless someone disputes them, and escalated to a token-holder vote if there's a fight. It's built for speed, global access, and permissionless participation.

These aren't cosmetic differences. Kalshi's structure assumes uncertainty should be contained: clear resolution criteria, bounded positions, participation filtered through compliance.

Polymarket's structure assumes uncertainty can be explored: fast iteration, unlimited size, let the market figure it out.

I'm the founder of Spredd Markets, a UI and routing layer built on top of Polymarket and Kalshi that brings prediction market access to ecosystems that aren't natively supported. Building that infrastructure means living inside the differences between these platforms daily. They're not variations on the same idea. They're different philosophies about what a market is for.

III. Where Structure Hits Price

The mechanics matter more than most people realize.

Take liquidity. Kalshi charges roughly 1% effective fees. Their market makers operate under compliance constraints and have to price conservatively in thin markets. The cost of being wrong isn't just losing money, it's regulatory scrutiny.

Polymarket charges zero trading fees. Liquidity comes from a different mix: AMMs, large holders, people chasing narrative momentum. It's cheaper to express a view, which means prices move faster. It also means they overshoot more often.

Then there's position concentration. Polymarket allows near-unlimited size. One trader can move an entire market. That's powerful when they're right and distorting when they're just well-capitalized. In a regulated environment with position limits, mispricing might persist longer, but corrections tend to reflect broader consensus rather than individual conviction.

Resolution trust is the other big variable. On Kalshi, you're betting the AP will call the election and Kalshi will honor it. That's a tight loop. On Polymarket, you're betting that UMA token holders will vote correctly if there's a dispute and there have been controversial calls. Traders price that ambiguity, which can widen distributions and sharpen tail prices in ways that have nothing to do with the underlying event.

IV. What the 2024 Election Actually Showed

The 2024 presidential election was the largest natural experiment prediction markets have ever seen. Over \$3.3 billion traded on Polymarket. Kalshi processed more than \$1 billion after winning their court battle to list election contracts. Same question, unprecedented scale, two very different machines.

I watched these markets daily. On election morning, Polymarket had Trump at 62 cents. Kalshi had him at 57. A 5-point gap on an identical binary question, one that should have been arbitrated away if these were really the same signal.

A Vanderbilt study by Clinton and Huang, looking at 2,500+ markets and \$2.5 billion in volume, found systematic differences. PredictIt with its \$850 position caps correctly predicted outcomes 93% of the time. Kalshi hit 78%. Polymarket came in at 67%. When they controlled for market types, the gap between Kalshi and Polymarket narrowed. Polymarket's lower accuracy partly reflected its appetite for niche, speculative markets that regulated platforms wouldn't touch.

The price behavior was more revealing than the accuracy numbers. Polymarket's daily moves barely correlated with Kalshi's. Fifty-eight percent of Polymarket's presidential markets showed negative serial correlation a spike one day, reversal the next. That pattern intensified in the final two weeks, exactly when prices should have been converging as information became clearer.

The Théo situation made the concentration dynamics concrete. One trader, \$42 million across four accounts, all on Trump. At peak, he held over 20% of all Trump YES shares. The largest Harris holder had 7%. When one person can move a market that much, you're not purely measuring collective belief anymore, you're measuring how much that belief was worth to someone willing to bet eight figures on it.

None of this means Polymarket's signal was wrong. It wasn't both platforms beat the polls. But the signals carried different information. Polymarket's price included the specific appetite of large, aggressive speculators. Kalshi's reflected a more constrained, compliance-shaped consensus. Both useful. Not the same thing.

V. Divergence Isn't Always Inefficiency

The instinct when you see prices differ across platforms is to assume someone's wrong. Arbitrage should close the gap.

Sometimes that's true. But arbitrage requires more than a price difference. You need to move capital between venues, handle different compliance regimes, trust that both platforms will resolve the same way. In practice, bridging funds from a Polygon wallet to a CFTC-regulated exchange isn't frictionless. The gap persists because closing it is harder than it looks.

More interesting: sometimes divergence isn't mispricing at all. It's two markets interpreting the same uncertainty differently. The regulated market prices institutional confidence, what the data says given formal resolution standards. The crypto market prices speculative conviction, what a particular group of risk-seeking traders believes will happen. Both can be rational in context.

When I'm building tools that aggregate prices across platforms, this is the core problem. Averaging Kalshi and Polymarket isn't just combining information, it's mixing two different kinds of signal. Sometimes the divergence itself is what you want to surface.

VI. Calibration Depends on Who's Trading

Most calibration discussions focus on whether the probabilities match outcomes over time. That matters, but it's incomplete. Calibration is also structural, it depends on who's in the market and what constraints they face.

Markets with position limits tend to compress tails. Extreme probabilities are expensive to express and hard to sustain when you can't size in. That leads to conservative early pricing and tight calibration near resolution. Markets with unlimited size express tails more freely, which can surface emerging belief faster but also amplifies overconfidence.

This is why a 5% price on Kalshi and a 5% price on Polymarket shouldn't be read the same way. One might reflect disciplined skepticism filtered through compliance constraints. The other might reflect aggressive pessimism from a concentrated position. Without knowing the structure, you're guessing at what the number actually means.

VII. How to Actually Use This

For anyone building on these markets or using them seriously, the practical question isn't which platform to trust. It's how to read each signal for what it is.

Regulated markets work better when you need resolution clarity, conservative calibration, or institutional defensibility. Crypto-native markets are more useful for early signal detection, narrative-sensitive domains, or situations where you care about what aggressive speculators think.

Averaging across platforms erases the distinction. Picking one as "the truth" ignores the other's information. The smarter move is treating disagreement as signal. When Kalshi and Polymarket diverge, that's telling you something about where uncertainty lives and how different participants are processing it.

I should be clear about what this paper can't tell you. The 2024 election is one event, and while it was massive, I can't prove these patterns generalize to sports or economic indicators or geopolitical markets. Both platforms continue to evolve. Polymarket has since acquired a CFTC-licensed clearing house and now operates legally in the US. UMA has moved to whitelisted proposers to reduce disputes. The structural gap I'm describing is already narrowing. And the

argument that divergence is “informative rather than inefficient” is genuinely hard to falsify. Maybe the gap is always just arbitrage waiting to happen. I don’t think so, but I can’t prove it conclusively.

Conclusion

If prediction markets are an information layer, then market structure is part of the message.

Kalshi and Polymarket aren’t just different implementations. They encode different assumptions about uncertainty, participation, and what makes a resolution legitimate. The 2024 election made this concrete: identical questions, different prices, different volatility, different responses to news and yet both beat the polls.

Treating their outputs as interchangeable obscures what makes them useful. Understanding the structure behind the price is what lets you read the signal clearly. These are instruments for navigating uncertainty, not oracles. The better you understand how they work, the more they can tell you.

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